Adithi Yellai

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.44
FIITJEE Junior College	Class XII (TSBIE)	2024	93.40%
The Future Kid's School	Class X (ICSE)	2022	97.00%

SCHOLASTIC ACHIEVEMENTS

- **Top 5** out of **23 IITs** in Adobe's ML problem statement in **Inter IIT Tech Meet 14.0.**
- **7th** out of **50+ submissions** in AI/ML paper presentation at Inter IIT Tech Meet **Student Academic Conference**.

PROJECTS

Gesture-Driven AI Photo Editing App (Inter IIT Tech Meet 14.0**)** (Link)

- Ranked **5th** in the Adobe ML challenge as part of a six member team.
- Developed a feature for an AI image editing application that **relights an image** under **variable scene illumination**.
- Integrated **stable diffusion** weights with custom UNet channel expansion for environment-conditioned generation.
- Reduced remote inference latency from 10s to **~6s** on 1024x1024 images using post-training **quantization.**
- Added mask-aware compositing, automatic shadow generation, and optional upscaling.

Semantic File Search Engine (Link)

- Developed a C++ based semantic file search application that crawls local directories, indexes file content and metadata, and stores vector embeddings in a **SQLite** database.
- Built core system components for crawling, tokenization, embedding generation with optional **ONNX** Runtime support, cosine-similarity, and file-change detection, also implementing CMake-based build automation and unit testing.

Image Restoration Pipeline (KLA Hackathon) (Link)

- Built a lightweight restoration pipeline that balanced denoising, deblurring, and super-resolution.
- Implemented a **SwinIR** transformer with a **hybrid loss function** combining FFT+SSIM+F1.
- Focused heavily on performance with **AMP** and system-level data ingestion workflows such as **DataLoader** enhancements (pin_memory, persistent_workers), resulting in a **45%** reduction in latency.
- Achieved 0.77 SSIM on validation set, a significant improvement over 0.54 baseline.

Compiler Infrastructure & Analysis (Compilers Course Project)

- Generated and analyzed **LLVM IR** using the LLVM/Clang 17 toolchain, examining how high-level C/C++ constructs are lowered into SSA-based intermediate representation.
- Implemented a lexical analyzer using **Flex** for a subset of the C language, supporting tokenization of identifiers, keywords, literals, operators, and comments with rule resolution.
- Studied compiler optimizations and IR transformations to understand control flow graphs, memory operations, and optimization techniques.

Protein Secondary Structure Prediction (Epoch Tenure Project) (Link)

- Built a novel TriESM-DSSP model for single-sequence secondary structure prediction by leveraging intermediate **hidden states** from ESM-2 and combining them through learned layer attention weights.
- Designed the data workflow to embed protein sequences, normalize and project layer outputs, and train a small **transformer-based classifier** for residue-level helices/sheets/coils.
- Validated the approach with layer-wise performance and mutation sensitivity ablations, achieving notable gains over final-layer-only baselines.

SKILLS

Languages: Python, C, C++, SQL

ML & AI: Numpy, Pandas, Matplotlib, PyTorch, TensorFlow, Scikit-learn, Keras, HuggingFace, OpenCV, MLOps

Tools & Infrastructure: Git, GitHub, Linux/Bash Scripting, Docker

RELEVANT COURSES

Computer Science: Algorithms, Data Structures, Operating Systems, Compilers, Theory of Computation, C Programming, Artificial Intelligence

Mathematics: Probability, Linear Algebra, Discrete Math, Statistics, Calculus

POSITIONS OF RESPONSIBILITY

Head of Epoch (AI/ML Club of IITH) :

- Designed a tenure plan, including budget management, industry collaborations, and community outreach initiatives.
- Recruited a team of 15 and directed their learning through multiple teaching sessions and curated tasks.

Core at Epoch (AI/ML Club of IITH):

- Delivered multiple machine learning sessions to the IITH community, each attended by over 100 students.
- Completed mini-projects and presented papers under Epoch during the tenure strengthen my skills.

EXTRACURRICULAR

Two-Time Inter IIT Sports Meet (Badminton)

Inter-College Sports (Volleyball): Represent IIT Hyderabad as a main player in inter-college volleyball tournaments

Creative Arts: Active member of the fashion club in IITH, contributing to creative design. Hobbyist painter